



Tourism recommendation system: a survey and future research directions

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Abstract

A Recommendation System (RS) is an intelligent computer based system which provide valuable suggestions to the user and are used in several domains. Social media platforms are the most common internet applications due to the large number of users. The numerous posts, likes, etc. have accrued on social media platforms and can be used in variety of recommendation systems. In this work, the primary focus is the tourism domain, where RS serves as a valuable tool for the tourist to plan his trip. Traditional RS systems only cater to the needs of the tourist by examining few factors. However, there are a large range of factors such as environment factors, actual geo-coordinates, trip destination, preferences of the user etc. that need to be taken into account in order to make a foolproof recommendation to the tourists. Tourism Recommendation Systems (TRS) provide suggestions to the tourists to identify the most suited transport (flight, train, etc.), accommodations, museums, special interest places and other items which are required for the trip. Several techniques are used and a thorough study of various techniques of traditional RS and TRS techniques have been done which are specially designed for tourism domain. Various Artificial Intelligence (AI) techniques have been highlighted which are used to solve the tourist recommendation problem. Also, future research direction has been suggested which would improve the Quality of Service (QoS) of the RS in tourism industry.

Keywords Tour recommendation · Social media · Tourism · POIs · Itineraries · RS · AI

1 Introduction

Recommendation System (RS) is an intelligent computer based system which serves as a guide and suggests, as per the preferences of the person [98, 163]. It uses state of the art

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techniques like Big Data, Artificial Intelligence (AI) etc., and benefits both the consumer and the merchant [93]–[114]. RS is becoming very popular as it serves as a guide, for the activity that a person or a group plans to perform in the best possible manner, given the constraints imposed by the user(s). In the tourism industry, a tourist has numerous options and can easily be lost. However, the RS comes to the rescue as a guide, by exactly suggesting about the various amenities that he/she needs to take in order to make the trip a memorable one [119]. RS have been present in the market, since the 1990's [87], and are constantly being upgraded by adding new techniques of machine learning and state of the art technologies like Smart Mobile Device (SMD), Big Data, etc. RSs are computer based intelligent techniques which solve problems of information over-loading, and are not search engines. RS is beneficial both to the consumers and the merchants and enhances user experience by assisting users in finding products information as per the taste and needs and also benefit the business by increasing the sales [97]. The key task involved in RS is to quantify how much a user likes an item and then predict the list of items that a user is most likely to buy. The approaches used in RS are either Content based filtering (wherein the system recommends the items that are similar to the items browsed by the user), Collaborative filtering and Hybrid filtering which is a combination of both the Content and the Collaborative based filtering. Various complex techniques such as heat conduction and mass diffusion are also being used in RS which have been introduced through advanced research done by physicists and complex system scientists [162]. An abstract structure of the social network involves different relationships between people, such as friendships, mutual interests and shared knowledge. Social networking services can be used to explore the actual social links between different users. It enables the users to exchange ideas, activities, likes, dislikes and news. These data can be used by different RSs for future recommendation of the items. Tourism Recommendation System (TRS) is a specific RS for the tourism industry and provides valuable suggestions and guidance to the tourists in identifying amenities (like transport, hotel, attractions and special interest places) as per the taste, interest, likes and budget to make the trip a memorable one. The various AI techniques used in TRS are Knowledge Representation (KR), Optimization techniques, Clustering, Multi-agent systems, etc [12]. The popular types of TRS are Web based and Mobile based, however, there are other types which need to be downloaded on desktops / laptops or on mobile devices and are not so popular, since tourists like to use simple systems without getting into much technical aspects. Web based TRS is commonly used in the e-Tourism industry as it presents a user-friendly interface providing images, maps, videos and good amount of data which the tourist can drill down as per interest. Mobile recommender systems (MRS) is gaining popularity as there are 7.2 billion mobile devices across the globe today, which is far more than the human population of around 7.19 billion [97]. A good example of MRS is a TRS which is installed on a SMD, and guides the tourist about the various necessities during the travel. Also, the cab drivers can easily find places and routes having less traffic so that the tourists need less time to travel to the places of attraction.

In this work, the authors have summarized the research on tourism recommendation system which will provide a strong foundation to the future or young researchers in the area of tourism domain. In true sense, the work will be helpful for designing a system which provides practical guidance to the tourist in order to make his/her trip a memorable one. The importance of tourism in today's world is immense as it is a big source of revenue and employment generation for a country. Hence, the economy of a country is elevated by the inflow of greater number of visiting tourist.

1.1 Motivation

The primary motivation for conducting this study is to explore prominent TRS developed so far and identify/highlight future research directions that help making TRS more robust, personalized and propose cost-effective suggestions to tourists. Tourism is a very popular industry with many advantages. It helps the country to earn revenue and also brings development in the particular area by promoting world class modern facilities to be present in that area [12]. Presently, the number of tourists has grown many fold, as people are becoming more global and exploring the world. Globalization of businesses has also boosted the tourism industry, as well as friends and relatives who are now spread in almost all corners of the world which has resulted in lots of travel by several persons around the year [12]. This phenomenal boost to the tourism industry demands a TRS to fulfill the specific needs of the tourist in a cost efficient manner. Tourism data can be taken from various Internet resources and travel agencies. Most of the travel agencies and Internet resources recommend popular attractions only. The recommendations may not fulfill the user expectation/interest. To overcome these problems, the current research community proposed numerous alternative approaches and techniques. Figure 1 shows the overview of the survey carried out in this work. The figure also shows that the work is mostly focused on the TRS. The work discusses the challenges, factors, features. The primary focus is on RS techniques that are used to handle tourist recommendation problems, as well as other strategies that are specifically built for the tourism domain. A number of AI techniques have also been highlighted that are being explored to solve the tourist recommendation challenge.

1.2 Contribution

The contribution of this work are as follows (see Fig. 2):

- A comprehensive overview of RS and their different categories are mentioned in this work and also, explore the usage of RS in TRS.
- The different functionalities of TRS have also been discussed here.
- The necessity of TRS is also presented in this study.
- Highlighted the use of a variety of social media sites in the field of tourism.

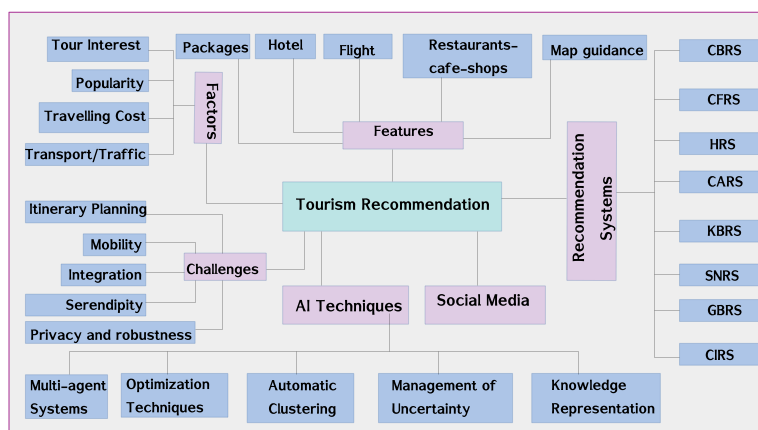


Fig. 1 Overview of the Survey

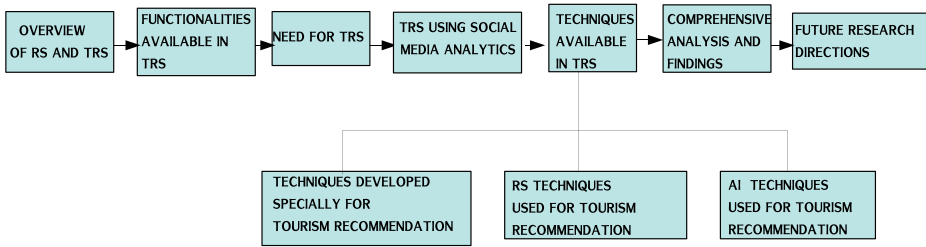


Fig. 2 Review methodology of TRS

- highlighted the tourism relevant problems which could be resolved by conventional RS. A comprehensive overview of the existing works which are specially designed for the TRS domain and also some of the AI techniques which are used in TRS are mentioned in this work.
- Categorize the discussed works based on the previous step.
- Towards the end, a few research directions have also been high-lighted with respect to the existing TRS techniques.

1.3 Article structure

The remainder of the paper is structured as follows, Section 2 discusses the comparison of related survey articles. Section 3 includes overview of RS and TRS. The functionalities available in TRS is presented in Section 4. Section 5 includes the need for tourism recommendation systems. TRS using social media analytics is presented in Section 6. Section 7 presents the techniques available in TRS. Future research directions are discussed in Section 8. Finally, Section 9 concludes the article.

2 Comparison of related survey articles

The work is different from the existing surveys [12, 87, 156] in many aspects (see Table 1).

(i) The earlier works discussed about the various AI techniques, which were used to solve the TRS problem. (ii) Existing survey works are mostly concerned with the tourism industry. Some of the recent surveys on TRS discuss about the personalized and group-based recommendation systems in the domain of Tourism. In this work, the primary focus is on the RS techniques which are used to solve the tourist recommendation problems and also some of the techniques which are specially designed for Tourism domain. Various AI techniques have also been highlighted which are used to solve the tourist recommendation problem. The main objective is to find the answer to the following question - Are the RS techniques good enough to provide recommendations to the tourist, or some other advanced techniques are required which may be used specially for the Tourism domain?. These answers can be obtained only after surveying the existing works in this area. The reviewed articles were gathered from many sources, including Web of Science, Google Scholar databases, ScienceDirect, and others.

The articles published in the field of tourism are reviewed before carrying out the literature review. Number of papers that offer different views on TRS were the outcome of this initiative (see Fig. 3) ([122]).

Table 1 Summary of the existing related works

Author	Contributions
[12]	Several features of these systems are examined, including their interface functionality, recommendation processes and AI approaches and techniques used.
[87]	This study examines current recommender system application advances and categorizes their applications.
[156]	Offer a brief assessment of the task of POI recommendation in Location Based Social Networks (LBSNs) outline potential research directions for POI recommendation

3 Overview of RS and TRS

This section covers the basic idea about RS and TRS. Recommendation techniques are mainly classified into Content Based Recommendation System (CBRS), Collaborative Filtering Based Recommendation System (CFRS), Knowledge Based Recommendation System (KBRS), Hybrid Recommendation System (HRS). However, recently there are other types of techniques used for recommendation, such as Computational Intelligence based Recommendation System (CIRS), Social Network based Recommendation System (SNRS), Context Awareness based Recommendation System (CARS), and Group Based Recommendation System (GBRS).

- 1) CBRS takes into account the content of products and helps the users by suggesting the appropriate item(s) depending upon the interest profile (U_p) of the particular user and the brief detail of the item. In CBRS, each attributes of an item needs to be compared and consequently, the best item is recommended to the user which has the high degree of similarity. The CBRS fails to provide the best suitable recommendations in case

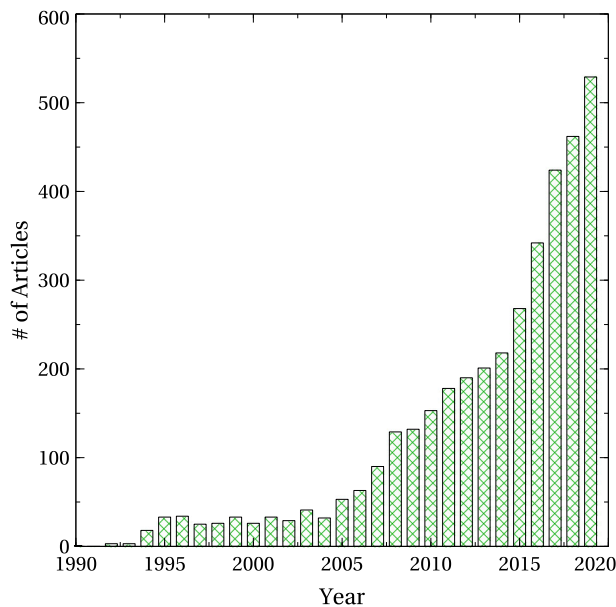


Fig. 3 Distribution of tourism recommendation system articles by years

sufficient information is not available about the user's likes and dislikes. U_p maintains information about the user interests and likes [110] which are detailed as follows.

- (a) There may be a large number of descriptions available about user's interest items but there needs to be a common representation function (f). For each item that the RS predicts there exists a likelihood that the use may be interested for the particular item.
- (b) A RS will contain the history of users' intersections which helps to recommend the particular item to the user with greater accuracy. For example U_p will be made when the user purchases any item (s) or the rating has been given for that particular item by the user. Also the history of the different types of queries will be stored.

The parts of U_p will be shown as a CBRS whereas Amazon has a feature named *favorite* which represents the details of each item categories which is preferred by an user.

- 2) CFRS is a useful technique used for deriving the recommendations which are suitable to a target user, and is based on the opinion of similar users having the same interests [105]. It is mainly of two types, memory and model based [131]. The model based CFRS has two different phases. In the initial phase, matrix containing ratings given by the user are used for generation of the model which represents the original matrix itself. subsequently, in the next step the model which is generated serves as an input to calculate the rating of predictions generated for a particular user. In the second phase of CFRS, the total matrix is used to calculate the prediction and can be split further into two approaches, which are *user based* and *item based*. In the user based approach, the recommendation is made on the basis of the preferences of another user who has similar tastes and likes. In case of an item based approach, the recommendation is made based on the user's past experience and preference of the particular item.
- 3) The user preferences are taken into consideration in KBRS and recommendation is provided based on the prior knowledge about the items, such that it suits the user's requirements. The knowledge is present in encoded form and serves as input data to the system. There are three varieties of knowledge which are fed into the system, namely, Catalog, Functional and User knowledge [17]. Knowledge Base (KB) is one of the key component of KBRS. Depending upon the quantitative and qualitative content of the KB, the KB is paired along with the recommendation strategy by involving similarity measures and matching techniques [17] and a U_p is created which is composed of the preferences of user, tastes, user interests, requirements, etc. The content of U_p depends on the methodologies as well as the strategy of recommendations.
- 4) It is also possible to build a RS wherein two or more different techniques can be integrated. In case of NewsDude, it normally uses the K-nearest neighbors (KNN) and Naive-Bayes classifiers for recommendation [10]. Burke et al. [17] describes seven different types of HRS techniques, namely Weighted [100], Mixed [130], Feature combination, Switching [10], Seature augmentation [106, 144] Meta-level, and Cascade [17, 109]. In the weighted technique, the votes or scores of different techniques used for recommendation are grouped in order to make a final recommendation to the user. In case of Switching technique, it is possible that different technique for recommendation are applied based on the present circumstances. In the Mixed technique, the recommendations are generated from different recommender's at the same time. In the Feature combination technique, the different features derived from various data sources involved in HRS, are aggregated to form a single algorithm. In case of Cascade technique, the output from a recommendation engine is refined by the subsequent RS,

thereby providing better recommendation to the user. In the technique, based on Feature augmentation, the output of an algorithm serves as the input for the next algorithm. In the Metalevel technique, a HRS is called Meta-level if the model is learnt using a particular recommendation algorithm and the output of the first algorithm is an input to the next one.

- 5) Dey et al. [29] has stated that “context” as the information required to define an entity. Entity can be an object (like person, place etc.) and is used for interactions with an user or an application

Based on a system which has full information, a RS can be categorized into three categories namely Fully observable, Partially observable and Unobservable.. In case of a RS which is of the type of observable fully, the value and the structure are explicitly known, A Partially observable RS has few contextual factors that are explicitly known.

- 6) CFRS offers more accurate recommendation to the users, however this technique suffers from various drawback like cold start, scalability and sparsity, which leads to inaccurate recommendations. In order to overcome these issues, it is suggested that CFRS be integrated with other computationally intelligence techniques such as Bayesian classifier, k-means clustering, Artificial Neural Networks (ANN), Fuzzy set techniques etc. and is called as CIRS. Bayesian classifier is used for solving the problem of classification which is based on the conditional probability definition as well as Bayes theorem and is useful for model-based RS. ANN consists of an assembly of nodes which are interconnected and the weighted links which are used for developing the model-based RS [4]. Hsu et al. [51], devised a Television RS using the ANN technique which uses the back-propagation NN (Neural Network) technique for training a NN which has three hidden layers. Clustering techniques are useful for grouping the similar items which minimizes the computational cost, like in the case of K-Nearest Neighbors (kNN) algorithm. G-r et al. [37], came up with a RS approach wherein the clusters were used for dealing successfully with the unrated data for each of the users. GA is useful for parameter optimization problems [59]. GA is mainly used for clustering [59], and hybrid user models [1]. A GA technique can be used for obtaining the function of the optimum similarity and it is clear that this function provides higher quality as well as faster results with respect to the traditional metrics. The fuzzy set techniques have been used when the necessary information about any situation or domain is incomplete.
- 7) The number of users who are using the social network is increasing day-by-day. Based on the analysis of social network, an intelligent RS can be made which can provide more interesting recommendations to the user. SNRS can provide the opportunity to users to interact with other users using the social network techniques such as social tags, comments, etc. SNRS fully utilizes the information available in social networks which boosts up the performance of traditional RS. The logic behind this technique is to utilize the simplified homogeneous social networks as a Bayesian network.
- 8) The large majority of RS are designed to make recommendations for individual users [166]. The items suggested to a particular user may not be intended for the single user, but may be recommended for a set of users. GBRS helps by providing useful suggestions to a set of users when members of a group are unable to interact face to face or do not have clarity about their preferences [41, 165]. GBRS is used in various areas like Entertainment and various complex scenarios.

The advantages and disadvantages of various Recommendation techniques are shown in Table 2.

Table 2 Summary of Recommendation Techniques

Techniques	Advantages	Disadvantages
CBRS [110]	• <i>User independence</i>: exploits solely the ratings which are given by individual user to setup their profile • <i>Transparency</i>: The recommended list of items that are provided by the CBRS are clearly explained based on the detailed working of the algorithm thereby providing sufficient explanation for the same. • <i>New item</i>: CBRS can provide recommendation even if the user has not provided any rating for the particular item. Hence the first-rater problem does not exist.	• <i>High scoring</i>: it suggests only the high scoring items which matched against U_p. • <i>Overspecialization</i>: there exists no inherent method in CBRS for finding any unexpected scenario. • <i>New user</i>: to understand the user preferences and make recommendations, system should collect enough ratings from the user. On the contrary, the system fails to provide suitable recommendations to a new user when less amount of ratings are available.
CFRS [131]	<i>Memory based</i>: • New data can easily be added • Implementation is easy • No need to consider content of an item • The scaling is done with respect to the corated items <i>Model based</i>: • Prediction performance can be improved • Addresses the issues related to sparsity, scalability, etc. • Provides an intuitive rationale for better recommendation	<i>Memory based</i>: • Determined by human ratings • Cannot provide recommendation for a new user • Limited scalability in case of large datasets • Performance gradually decreases when data is sparse <i>Memory based</i>: • Trade-off between prediction performance and scalability • Useful information can be lost due to the application of dimensionality reduction techniques • Expensive to model
HRS [10]	• Overcomes the drawback of CBRS, CFRS and other RS by integrating two or more RS • Prediction performance can be improved • Overcomes sparsity and gray sheep problem	• Due to the integration of two or more RS the complexity and expense may be increased • External information is required which may not be available
CARS [29]	• When the user interacts with a particular context, this technique helps to resolve the problem of “situated actions”	• Most of the techniques used in CARS center around the context, whereas the other alternative methods are under-explored • Prefiltering, postfiltering, and contextual modeling paradigms are partially explored
KBRS [17]	• Can map the user requirements to products • KBRS include non-product features as well • Ramp-up is not required	• The capability to learn on its own does not exist • There is a need for knowledge engineering to be incorporated in KBRS

Table 2 (continued)

Techniques	Advantages	Disadvantages
SNRS [159]	- Improves the accuracy of recommendations which are dependent on the social interaction pattern of users - There is no need to find out the similar users	- user who have less interest in networking with others do not contribute significantly in such type of RS - Recommendations provided to the users who are not active in social networking may not be accurate
GBRS [165]	Solves the cold-start problem, wherein a suitable recommendation can not be provided due to lack of information	There exists very large degree of complexity to provide recommendation to the group of users
CIRS [51]	Overcomes the issues related to CFRS like cold start, scalability and sparsity	- Dealing with uncertainty - Most of the time computational intelligence techniques is very complex to design and can only be handled by the expert as black box approaches - Over-filtering wherein the performance is good for training dataset but poor performance for the test dataset when a model become very complex.

3.1 Tourism recommendation systems

TRS provides suggestions to the tourists to identify the most suited transport (flight, train, etc.), hotel, restaurant, map guidance and other items which are required for the trip. The recommendation for the trip and planning of itinerary are difficult tasks for the tourist. Point-of-Interest (POI) is an important parameter to design a TRS. POI refers to a definite location point which the user finds interesting in some aspects [33]. There are various factors namely tourist interest on a particular POI, popularity of POIs, time or distance travel costs, transport/traffic conditions etc. which play an important role during recommendation of POI.

A personalized POI consists of two basic elements such as the information related to the particular POI and the user [61, 155].

- 1) *Attributes related to POI*: The information can be split into subjective and objective attributes. In case of subjective attributes, the information is collected using the visitor's opinion for a particular POI, whereas, the objective attributes include the information about the tourist such as location of POI [11].
- 2) *User attributes*: The user attributes affect the POI selection which consist of user's personality, behaviors, social factors, etc. A particular user may tend to select different POI with respect to another similar user, even though the same tourist information is presented to two different users.

The attributes can be further split into static and dynamic types. The static attributes need input data only during initial registration. In the dynamic attributes the user's input is considered at all times. The recommendation is continuously changing because of the changing condition of the user during a tour [33].

The POI recommender systems are different from the traditional RS in some aspects which are detailed as follows [156].

- 1) *Geographical Influence (GI)*: This particular characteristics of GI distinguishes the traditional RS and POI recommender system. In general, a user may be interested to visit nearby places which indicates that a user may visit several POI's surrounding a particular POI.
- 2) *Frequency data and sparsity*: In traditional RS, a user generally expresses his/her interest using ratings, where a higher rank indicates higher satisfaction level and may form User-item Rating Matrix (URM). Unlike to traditional RS in POI recommender system, a user's preference is made based on his/her frequency of visit to various locations which is termed as *User Location Check-in Frequency Matrix (ULCFM)*. The sparsity of ULCFM is larger than URM.
- 3) *Social Influence (SI)*: To increase the recommendation quality, traditional RS combines the social relationship between friends with ratings, but in [48], the authors have shown that, 96% of users share less than 10% of commonly visited places of interest.
- 4) *Temporal influence*: In case of POI recommendation, temporal influence plays a vital role because during the check-in activity, there might be physical constraints as well. Generally, in case of POI, there are mainly three aspects of temporal influence, namely, consecutiveness, non-uniformness, and periodicity [38]. The behavior of user's check-in exhibit a periodic pattern because a user may check-in restaurants at noon but also have fun at nightclub. On the other hand, the non-uniform aspect of temporal influence depicts a user's preferences of check-in which may vary depending on the specific hour of the day, specific month of the year, etc.
- 5) *Content Indication*: A user may generate contents, including POI ratings, geo-tagged photos, etc. [112]. It is possible that the related content might not be tagged to each of the check-in record. The user comments may be used to improve the degree of POI recommendation.

A proper recommendation can be made using a suitable *itinerary planner* which is basically a computer software program. The itinerary planner is completely different from the exiting trip planner in three aspects [78]. Firstly, the itinerary planner has considered multi-modal itineraries, secondly, the itinerary planner has ability to handle multiple constraints and also the time windows for arrival as well as departure time. Finally, in order to recommend the best itinerary for a particular user, the itinerary planner should adopt an interactive programming approach. Moreover, to designing an itinerary planner, there are two challenges. Firstly, on which criteria itineraries can select by the planner. Generally, in case of real-world situation, a single itinerary is not sufficient and to select itineraries, a planner must consider multiple criteria and trade-offs among them. On the other hand, the second challenge is constraints which are associated with the trip. These constraints are imposed by travelers or exist as intrinsic attributes of a particular visit.

For modeling personalized tourist guide, there are three functions that need to be considered, namely, the profile of a tourist, route generator, and finally, customization [78]. In addition, the tourist factors are dependent on the different characteristics such as social-demographic, geographic, etc., and tourist's preferences such as POI, time and money budget, etc. [47] discussed about the Tourist Trip Design Problem (TTDP). There are three methods available for TTDP, namely, the profitable Tour Problem (PTP) which maximizes the profit. The Prize Collecting Traveling Salesman Problem (PCTSP), minimizes the

traveling cost with profit of the total tour and is not less than the actual value. The Orienting Problem (OP), maximizes the profit, keeping travelling cost below a certain value.

Generally, TTDP problem involves with problem of the route planning in case a tourists plans to visit multiple POI.

When the tourist wants to visit multiple places which are his/her POI, the route covering all places needs to be planned efficiently, such that he/she is able to cover all places in an orderly manner keeping in view the constraints such as time required, distance between the different POI's, cost, etc., and also ensuring that the satisfaction is maximized and the trip is really a memorable one at the end of the day. This aspect of the trip is termed as the TTDP and involves intelligent algorithmic approach and methodologies in order to resolve the trip design issues using the constraints of the tourist and the different attributes of the POI. The new prospects in trip planning problems are the arbitrary start / end tour locations, solving realistic TTDP formulations, fast tourist routes updates and parallel computation.

Keeping in mind, the various constraints of the tourist and the POI attributes, ordered visits are planned by the daily tourist tours using TTDP. The main aim is to select a POI which matches the preferences of the tourist leading to maximization of a satisfaction of the tourist, and also taking into account the various parameters and constraints. The aim of this work is to survey the various models, algorithm and methodologies which are related to the tourist trip design. This work also examines the recent approaches, processing on the models which can capture a multitude of realistic POI attributes in the best possible manner. Different open issues and the promising aspects of future research work have also been discussed.

3.2 Summery and insight

Tourism is a kind of activity that strongly connects tourist's interests and preferences. In any case, it is difficult to apply RSs to the tourism domain. A lot of components must be ranked in order to apply CBRS (see Table 12). Generally, a tourist can't rank a lot of components. CFRS is a helpful method used to derive suggestions that are appropriate for the target user and is based on the view of comparable customers with the same interest. However, CFRS can not make a suggestion for a new tourist. It is also feasible to construct a RS in which two or more distinct methods can be incorporated. Due to the inclusion of two or more RS, the complexity and expenditure may be improved. On the basis of an assessment of the social network, a smart RS can be produced, which can make more interesting suggestions to the customer. Users who have less interest in networking with others do not make a significant contribution to this sort of RS. Generally, a person travels with different people, so the inclinations of a couple of tourist ought to be borne as a main priority. GBRs helps to provide helpful suggestions to a number of tourists. It is evident from the discussion in Section 3 that the traditional RSs are not adequate to fix the TRS issue. Therefore, there are so many works specially intended for tourism domain.

4 Functionalities available in TRS

There are various functionalities available in TRS such as Trip Planner, Social Aspects, Tourist Packs and Suggested Attractions [120]. These functionalities are discussed below.

- Tourist Packs: Seidel et al. [127] provides recommendation regarding the tourist destination and purchase of different services such as booking of the trip from their personal

travel agents. Lorenzi et al. [85] used travel agencies to give best packages on the basis of user preferences.

- **Suggested Attractions:** TRS suggests numerous restaurants, hotels, attraction location, etc. to the tourist when he/she has already fixed the destination point or availability places. Ruotsalo et al. [118] proposed SMARTMUSEUM which can easily detect when a user is present either indoor or outdoor. Gavalas et al. [46] developed a TRS based on collaborative filtering techniques by considering the user's current location, weather condition and also the already visited places. The authors have installed Wireless Sensor Networks (WSNs) on the user side for uploading user ratings. [83] analyzed the chat messages between the tourist and the travel agent. The system automatically searches the database and retrieves the user specific elements such as user attraction or cities based on their preferences.
- **Trip Planner:** Lot of work has already been done provides list of places from which the user selects their target places according to their needs and also to creates a route combining several attractions. Huang et al. [55] proposed an ontological based Formal Concept Analysis (FCA) approach in order to integrate the heterogeneous tourist information for online tour planning.
- **Social Aspects:** A good amount of work has been done based on the social functionalities such as picture, comments or videos etc. and interaction with other users. Kosmides et al. [63] developed a recommendation technique based on machine learning. Authors have used the social networks data in order to produce best recommendation system for the user. In [57], the authors have used the information of the social network in order to improve the Top-k recommendations and has been improvised using the Matrix Factorization and Nearest Neighbor algorithms [150]. In [138], the authors have proposed a tourist recommendation system named as GuideMe which integrates the social networks and the unique set of options which are given in the applications.

5 Need for tourism recommendation systems

In case of TRS, the main actor is a tourist who is exploring new places. Tourism is mainly classified into various categories such as medical, educational, etc. all these domains consist of a large number of information which are stored digitally. This work addresses various technical challenges where traditional RSs are not sufficient to solve the problem of TRS.

- *Dynamic Itinerary Planning:* To plan an optimal itinerary is a great challenge for tourists. Generally, a tourist may visit different places in a city, restaurants etc. The existing RSs recommend tourists based on tourist's interests, but their nature are static which means, the systems ignores the real-time changes of different places.
- *Mobility of Users:* The prevalence of mobile technology in the world has boosted the popularity of Mobile RS (MRS). Today, SMD will be present in every human being's hand, and often more than one for a single person. Life has become unimaginable without a SMD for people across the globe today. The advantage of using MRS is that the tourist can use it anywhere as SMD's have an internet connection and access to information is there at all times, which provides up to date suggestions to the tourist keeping him abreast of the situation of the particular amenity or the attraction. Also, the SMD's have GPS which provides Geo-referenced information and guide the tourist to the right attraction as per his interest. It is also necessary that all relevant information is squeezed

- in the well-structured manner onto the screen of the mobile device (whether it be a mobile phone, tablet, laptop, etc.) to help the tourist in fast manner.
- *Evaluation Methods*: Most of the techniques for RS consider tourist's feedback for evaluation methods using Mean Average Error (MAE) and Root Mean Squared Error (RMSE).
 - *Integration*: The main challenge in TRS is the seamless integration between the various sources of information such as airline, restaurants, and travel etc. It is required to integrate this information in a condensed format. For example, when trying to extract the information about the preferences of individual tourist, it is necessary to get the information from various systems. The integration plays a mazure role in this case.
 - *Serendipity*: Serendipity is the concept of exploring a new interest which the tourist has no idea about. For example, a tourist having interest in art history then the same tourist might also have interest towards ancient monuments. This type of model can be derived by the usage of intelligent ML algorithms, which processes the huge amount of tourist data.
 - *Privacy and robustness*: Privacy and robustness are a great challenges in case of TRS domain. It is very essential to protect the privacy as the systems consist of large amounts of personal information. It is also very important to protect the systems from different kinds of attacks.

6 TRS using social media analytics

Social media platforms like Twitter and Facebook, and others, such as Flickr, can be used to store, search and upload images or videos [78] (see Fig. 4). Foursquare App offers personalized advice for locations near a current user location based on user background, transactions and check-in background [107], etc. Few works have shown that the recommendations can use Yahoo Maps, Google Maps and OpenstreetMaps, etc. [116]. Some other researchers [78], have used by Wikipedia to extract POIs, latitude/longitude coordinates of POIs and categories of user interest. Rashid et al. [113] present a thorough literature assessment based on 109 high-quality research publications selected after a rigorous screening approach. According to the statistics, there are eight basic categories of social media intention. A taxonomy of the tactics and approaches used for intention mining is also included in this work. The social media intention may be used to determine tourist intentions, transportation plans, and involvement in online travel businesses. This can further be used to create a tourist recommendation system.

TRS requires two kinds of information namely, information about users' preferences and tastes and information about different products. Both can be obtained from the social media. Social media may be used to detect new products or to apply more characteristics to existing product in the social media database. Moreover, the social media concentrates on the identification of behavioral patterns, desires, and personal characteristics of users.

There are few techniques which are used in the recommendation process from RS such as CBRS, CFRS, KBRs, HRS etc. But due to the rapid development in the tourism industry, these techniques are not adequate to provide recommendations. Therefore, a few techniques have been specifically designed for the TRS domain. A few techniques have also been developed using AI techniques to provide better recommendation.

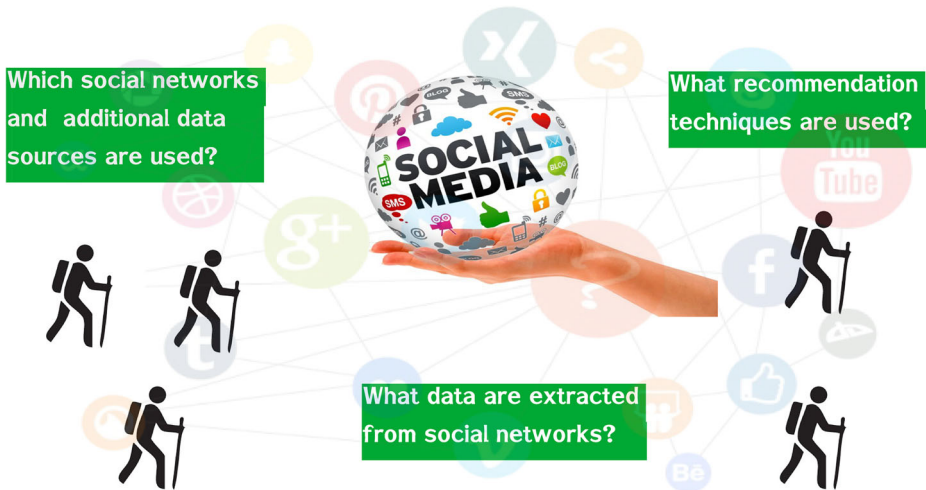


Fig. 4 TRS using social media analytics

7 Techniques available in TRS

TRS techniques can be categorized in following ways. First, some of the RS techniques which are used to design TRS have been highlighted. Then the techniques which are specially designed for TRS domain have been presented. Finally, the TRS techniques which are developed based on AI techniques have been explained.

7.1 TRS techniques from RS

Garcia-Crespo et al. [44] discussed about Sem-Fit which is a semantic TRS, recommending hotels to tourists. It helps to reduce the cost incurred by tourists in order to search for accommodations, their location, and other amenities based on the taste of the tourist. It uses fuzzy logic and ontologies of the hotel industry domain. Linaza et al. [80] presents a TRS which generates personalized trips for the tourist along with the images of the attractions. Details of the tourist are captured and used by TRS to present the list of tourist locations as per his interest.

Lorenzi et al. [85] presents a TRS, named PersonalTour which suggests best travel packages to the tourist, and is based on the concept of Distributed AI wherein a single recommendation request is split into partial recommendations handled by different individual agents. Montejo-Raez et al. [101] presents Otium, which is a planning system that helps in scheduling the tasks of a tourist by creating an agenda within the tour dates by considering the user's taste and other parameters. In Fig. 5, there are two different kinds of databases, namely, *profiles database* which is used for storing the user preferences and *event database* which is responsible for storing various events such as sports show, dance show, etc., The web interface which is Ajax based, helps to increase the system performance. Ultimately, *event database* is updated using an off-line process which gathers the various events

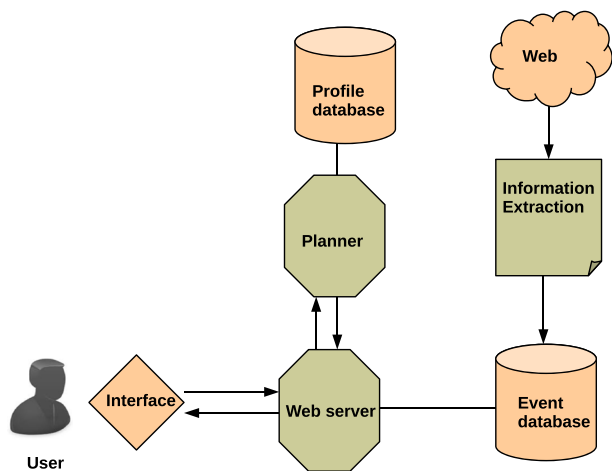


Fig. 5 Otium Architecture [102]

using web services. Figure 6 represents the CORS which consist of three main components, namely Recommender, Constraint Generator and Constraint Solver. A Recommender takes input from the user’s rated component. A Constraint generator takes the constraints such as travel distance, budget, etc, as input. A Constraint solver works as an optimizer of both recommender and constraint generator. The recommendation component takes input such as trip to a country, tourist attraction, etc, and produces a suitable configuration. An example of KBRs system which is used in TRS has been presented in Fig. 7. In Fig. 7, the client requests the server for various information such POI, maps, etc., which depends on various factors such as the actual geo-coordinates, trip destination, and preferences of the user. The user interface provides a huge label of interaction with the users. The planning engine component is used to generate the plan. Map Server (MS) acquires the map from external server and subsequently, the route server calculates the route from the Google map. Ge-Ontology

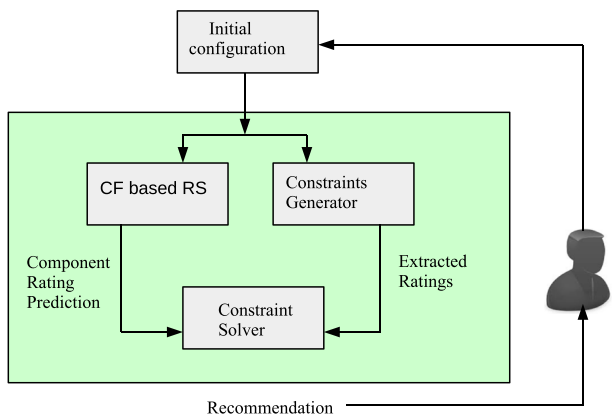


Fig. 6 Complex Objects Recommender System (CORS) [121]

7.2 Techniques available in TRS domain

Some researchers are focusing on developing different techniques which are associated with the traveler behavior (see Tables 3, 4, 5 and 6). For instance, MinFeng et al. [99], developed an isotropic dynamic spatial lag panel origin eDestination (OD) travel flow model, in order to comprehend the travel pattern of the Australian domestic and international travellers. In spite of these works, there are few challenges which need to be understood and captured fully. Few of the points regarding the behavior of international travelers are highlighted below.

- 1) *Capturing of tourist data:* Survey and opinion poll constitute the most common mechanisms for understanding the travel behavior of tourist. However these methods are time consuming and many of the tourist may not respond and also, the data might not reflect the actual travel pattern
- 2) *Travel Preferences:* There is a common assumption that the different tourist group exhibits different travel behaviors [78]. Tourist belonging to different countries may have different preferences such as their staying period, types of travel attractions, etc.

Li et al. [72] provide a consistent technique for evaluating context-aware similarities across distinct users by examining the influence of temporal and geographical aspects on the users. They offer a method for creating dynamically unique POI suggestion lists for each user based on the user's current context information. Sarkar et al. [120] provided a TRS framework for recommending multiple itineraries. The authors have also presented a framework in which multiple itineraries were suggested based on POI within POI [119].

Recently the advantages of mobile technologies, multimedia [146] and also the popularity of social media have boosted the growth of huge amount of data generated by the user such as photographs of different places with built-in GPS technology. The exact geographical location of the traveler can be obtained from the latitude, longitude, and other metadata of each of the photograph. The geo-tagged data helps to extract the necessary information such as the tourists' travel behavior, travel pattern, tag completion, POI extraction, etc. Yin et al. [152] described an approach for mining the trajectory patterns which are frequently visited using the geo-tagged collection of different pictures. The authors suggested that the GPS locations need to be identified before extracting trajectory pattern.

In a single day, the trajectories of a particular tourist are created depending on his/her visit to the different POI. If a trajectory pattern is repeated frequently, it is considered as a pattern of frequent trajectory. Geotagging includes attaching the GPS coordinates to the photographs using the Exchangeable Image File Format (EXIF) [35]. During the process of Reverse geocoding, the name of the place from the photograph. A reference database based on geospatial data is necessary to extract different parameters. This process is useful to identify the user's POI but may not be sufficient enough to mine the necessary data [35].

Liu et al. [81] proposed the geo-tagged photo data set based approach which divides the classification process into three parts namely, city, important places and tourist behavior. Ickjai et al. [56] proposed a mining framework for POI, taking the Flickr data set into consideration. The data is then further preprocessed in order to ensure that the duplicates data does not exist and clustering algorithm is subsequently applied. The multiple photographs taken by user at a particular location are considered for the purpose of a single case study. Furthermore, to identify the clustered POI pattern, the preprocessed datasets are fed into the clustering algorithm.

Table 3 Summary of Techniques which are available in TRS domain

Authors	Summery	Disadvantages
[115]	– Successive time-stamp for recommendation of the attraction. – Personalized recommendation problem was resolved using the factorizing personalized Markov chain (FBMC) method.	This work does not truly reveal the real-time situations of a particular attraction.
[164]	– Spatial-temporal latent ranking model to recommend the best possible successive attractions considering the time significant.	This work does not take into account the personalized factor and also does not truly reveal the real-time situations of a particular attraction.
[73]	– time-aware factorizing personalized Markov chain (TA-FBMC) model. – 4-dimension static tensor in order to store both the temporal and spatial check-ins data. – Time-decay factor by calculating the difference between two successive check-ins.	Complex model wherein every calculation uses all the check-ins data. Also, the authors have taken the user as an extra dimension which is Redundant. Hence, in case a new user is included in the RS the performance is degraded.
[74]	– Completion algorithm based on fast low-rank dynamic tensor. – Recommendation using the pigeon-inspired heuristic algorithm.	There exists an issue wherein the recommendation is not matching to a particular tourist as the data is taken from the users who do not belong to that particular location. This problem exists when there are a large number of outsider in a particular place.
[159]	– Generate the entire package for the tourist which contains the attractions and the optimum sequence in which the tourist needs to visit them. – Uses the collaborative filtering technique to determine the preferred attraction of the particular tourist.	A specific POI has not been taken into account.
[151]	– Used Spatial-Temporal LDA (ST-LDA) to come up with an efficient attribute pruning algorithm.	In case the tourist are on the same social networks there interest will be same.

After the clustering process has been completed on the data of the tourist attractions, the identified clusters can be assumed to be as POI. In practice, a single cluster is formed using the geo-tagged photographs of several POI. Consequently, there will be many POI which may not have been mined. To resolve this issue, T-h et al. [134] came up with a novel framework for POI mining which is based on clustering at two levels namely, random walk and constrained clustering. The authors had separated out large amount of photos which were geo-tagged into several groups in the case of random walk clustering. On the contrary, constrained clustering divides the clusters which are associated with several POIs into sub-groups.

Table 4 Summary of Techniques which are available in TRS domain

Authors	Summary	Disadvantages
[2]	– Probabilistic model in order to established the relationship between user-annotated tag and the attraction like-ability. – Scores are computed from multiple LBSNs and integrated using learning to rank techniques.	The link between a particular season of the year and the specific attraction has not been done appropriately.
[152]	– Mining the trajectory patterns which are frequently visited using the geo-tagged collection of different pictures. – In a single day, the trajectories of a particular tourist are created depending on his/her visit to the different POI.	The POI recommendation made are not at all diverse in nature.
[56]	– Mining framework for POI, taking the Flickr data set into consideration. – To identify the clustered POI pattern, the preprocessed datasets are fed into the clustering algorithm.	Many of the attractions are not mined using this particular technique.
[160]	– Manual investigation on real-world datasets	The link between a particular season of the year and the specific attraction has not been done appropriately. It does not recommend the attractions which are time-based by taking into account the duration of the user's visit to the specific POI.
[90]	– Co-pairwise ranking model by assuming that the POIs which are near to the user's previously rated POI are given a higher rank. – Understanding the different context such as time, date, weather, collective wisdom of users, etc.	The recommendation does not take into account the similarity of users, but on the contrary, uses a simple similarity computation to make the recommendation.
[28]	– Guessed the favorite locations of the users from Flickr photos by measuring the similarity between the different tourists on the basis of travel histories and similarity values.	It does not recommend the attractions which are time-based by taking into account the duration of the user's visit to the specific POI.

Table 5 Summary of Techniques which are available in TRS domain

Authors	Summary	Disadvantages
[76]	– Developed PERSTOUR algorithm to recommend tours which are personalized on the basis of popularity of the POI and preferences of the user.	The exact calculations of the arrival and departure time is not possible through the geo-tagged photograph that the tourist uploads.
[65]	– Recommends a travel route for the tourist based on the geo-tagged and time-stamped photographs held by Flickr. – Depending upon the photographer's behavior, a photographer behavior model has been developed.	It does not recommend the attractions which are time-based by taking into account the duration of the user's visit to the to the specific POI.
[77]	– Recommends a travel route for the tourist based on the geo-tagged and time-stamped photographs held by Flickr. – Suggested a solution on the basis of clustering for a group of members, for recommending POI and finally, for assigning guides using the technique of integer programming.	The exact calculations of the arrival and departure time is not possible through the geo-tagged photograph that the tourist uploads.
[64]	– Recommend a tourist based on data from the internet sources like social media and various other websites. – Used the technique of Collaborative filtering in order to determine few of the attraction.	During the tourist's visit to several attractions the context information is not considered.
[22]	– Used the Chebyshev polynomial approximation method to derive the tourist preferences based on the location and details present for that particular location.	The link between a particular season of the year and the specific attraction has not been done appropriately.
[153]	– Time-aware metric embedding algorithm wherein the factor of the user visiting the attractions at different point of time is taken into account. – Dynamic approach.	The link between a particular season of the year and the specific attraction has not been done appropriately.

Table 6 Summary of Techniques which are available in TRS domain

Authors	Summery	Disadvantages
[14]	– Personalized Touristic Tour (PTT) framework and modeled after the Generalized Maximum Coverage problem. – Unsupervised method for ex- tracking the tourists movement pattern in a particular area. – Helps to suggest the best possible trip within the constraint of the travel time of the tourist. – Explained different methods for personalized POI recommendation such as computing algorithm of the similarity, the first POI selection and final POI recommendation method.	It is not possible to calculate the transit time between the different attractions visited by the tourist from the geo-tagged photo. The link between a particular season of the year and the specific attraction has not been done appropriately. The specific attraction has not been identified appropriately with respect to the particular time duration (e.g, duration of a winter or summer season).
[86]		
[34]		

Sieh et al. [129] proposed a time-sensitive route RS named as TripRouter by considering the popularity of places, order of visiting of different places with time and also the proper timings for transition from one place to another. The authors have considered two major functions as follows.

- Developing the route using the source location of the user and the starting time.
- Compose the path at a particular starting time between source and the target location.

Yuan et al. [160] conducted a manual investigation on real-world datasets and found that, around multiple centers, each of the traits is likely to cluster around multiple centres. As a consequence, the authors have proposed co-pairwise ranking model by assuming that the POIs which are near to the user's previously rated POI are given a higher rank. The proposed method can have a greater liking for one alternative place over another from non-observed rating pairs, and also tends to alleviate the problem of sparsity of matrix factorization.

Majid et al. [90] devised a system that mines the POI of the tourists and sequences of travel from the geo-tagged photos by understanding the different context such as time, date, weather, collective wisdom of users, etc. De C. et al. [28] extracted the Flickr photographs and accumulated them into a graph and subsequently given different types of itineraries based on the availability of the users. Cheng et al. [23] came up with a RS algorithm based on the particular profiles of the user such as age, gender etc. Clements et al. [24] guessed the favorite locations of the users from Flickr photos by measuring the similarity between the different tourists on the basis of travel histories and similarity values.

The geo-tagged social media data helps in identifying the various tourist's landmarks as well as the tourist hotspots using various clustering methods. Brilhante et al. [14], [15] proposed *TripBuilder*, which is a Personalized Touristic Tour (PTT) framework and modeled after the Generalized Maximum Coverage problem. The authors have considered Flickr for data of the original itineraries of the tourists and matched them with the touristic POI from Wikipedia. Also, the authors have introduced an unsupervised method for extracting the tourists movement pattern in a particular area. This method stores the tourist's POI, popularity, visiting times, etc. Lu et al. [86] proposed Trip-Mine which is developed based on the data mining based approach. The work helps to suggest the best possible trip within the constraint of the travel time of the tourist. The authors have claimed that this works is the first work which helps to make an efficient plan within the travel time constraints.

Souffriaue et al. [132] came up with an approach which helps the tourist in taking fast decision using the mobile device. The authors have solved the Tourist Trip Design Problems (TTDP), which takes into account a set of location point having a particular score and the aim is to increase the cumulative score of the location where the user has visited. On the basis of the vector space model, the scores are calculated which represents the interests about the particular location.

Vansteenwegen et al. [140] proposed an approach for Mobile Tourist Guide (MTG) which acts as a user-friendly guide for the tourist. The mobile device can immediately recommend a holiday plan on the basis of real-time data. The authors have considered the TTDP problem as an extension to the Orienteering problem. E-y et al. [34] have explained different methods for personalized POI recommendation such as Computing Algorithm of the Similarity, the first POI selection and final POI recommendation method. Lim et al. [76] developed a model of tourist interests over time by considering his past visit to the POI and the geo-tagged photographs. The author has developed PERSTOUR algorithm to

recommend tours which are personalized on the basis of popularity of the POI and preferences of the user. The following steps have been suggested for providing the best recommendation to the tourist.

- Initially, the POI visits determined based on the geo-tagged photos.
- A travel history is constructed by storing the tourists' POI visits list.
- Recommendation is based on the PERSTOUR.

Kurashima et al. [65] came up with a technique which recommends a travel route for the tourist based on the geo-tagged and time-stamped photographs held by Flickr. Depending upon the photographer's behavior, a photographer behavior model has been developed, which consist of two models namely, topic model which is responsible for calculating the user's individual preferences and the Markov model which is responsible for finding out the routes of the photographers. Lime et al. [79] introduced a problem GROUPTOURREC which addressed many challenges such as forming a group of members who possess similar interests, recommending POI. In order to resolve this problem, the authors suggested a solution on the basis of clustering for a group of members, for recommending POI and finally, for assigning guides using the technique of integer programming.

Maruyama et al. [94] proposed p-Tour, an individual navigation device which calculates the route of the tourist. The author, have used a variant of the TSP with profits which focuses on determining a circuit which will reduce the cost after subtraction of the profit collected. Yu et al. [158] tried to solve a personalized tour recommendation travel package in order to predict an attractive tourist plan. The authors, have considered three problem statements, like which place should I go? How long can I stay in a tourist place?, and what should be the appropriate sequence of visiting the places?. Authors have modeled user references, location popularity, and location dependence. To discover the user's POI, the authors, have divided the total duration of travel into small time intervals and select the POI's from a set of POI locations for planning the route and compute the rank on the basis of user preferences. Zheng et al. [165] came up with two types of recommendation engine based on mining the GPS traces of different users. The first approach recommends based on the locations which are on top of the list of interests of the particular user, by modeling a Tree-based Hierarchical Graph (TBHG), consist of user's location history. The second one is a personalized recommendation which provides a personal with matching of their locations her travel preferences.

Yuan et al. [161] suggested a framework named as DRoF which helps to discover the different regions of a different functions in a city by considering mobility of the users in that region and their POI. The author, have divided the city into different segments such as highway and urban express way, etc. and using a topic-based inference model, authors have inferred the functions (POI's categories, restaurants, etc.) of each region. Uddin et al. [137] showed how to measure the Region of Interests (ROIs) in databases. ROIs defined as the region where the number of moving objects are present for a given time interval. The proposed approach does not depend on the ROIs parameter, which helps to identify ROIs very fast without rescanning the entire database. Yoon et al. [154] proposed recommendation techniques for large itineraries which are efficient and balanced using multiple GPS trajectories. To receive an itinerary recommendation, user needs to generate a small query, a finishing point and, also the travel duration. Lu et al. [88] tried to solved automatic travel route planning problem for tourists. The authors collected around 20 million geo-tagged photographs from *www.panoramio.com*, which helped in coming up with a route plan based on the user's preferences. Liu et al. [82] analyzed the different characteristics of the

various travel packages and developed a model, named Tourist-Area-Season Topic (TAST), which extracts different topics based on the intrinsic features of the landscapes. Clements et al. [24] came up with a model to recommend tourist location using the tourist's visiting history. De et al. [28] developed an approach which automatically constructs intra-city travel itineraries based on geo-temporal breadcrumbs which are left by the tourists.

Chih et al. [135] came up with a personalized RS which helps to provide suggestions about a sequence of visiting spots when the users takes the photo of a current scenic spot. The authors have used data mining techniques to extract and distinguish the various users' preferences. Ashbrook et al. [5] discussed about location context which helps to develop a predictive model of the tourist's future movements. The authors, have proposed a system which automatically clusters the user's GPS data which can be taken over an extended period of time. Liliana et al. [75] proposed INTRIGUE which is a prototype tourist information server. It presents the necessary information about different areas of Torino city. INTRIGUE helps to recommend sightseeing destinations as well as itineraries keeping in mind the preferences of heterogeneous group of tourists and group member's requirements. Maruyama et al. [94] came up with a Personal Navigation System (PNS) which helps a tourist to navigate through multiple destinations. PNS computes the route including user's destination by satisfying the requirements of the tourists and navigates the user. Wu et al. [145] formulated the problem for scheduling the plan based on the weather condition. The authors, have composed schedules for changing the weather probabilistically when the future weather probability has been given. Hu et al. [52] proposed a Stamina-aware Sightseeing Tour Problem (SASTP), which says that, a tourist can propose various sightseeing spots to be covered in a certain time interval and stamina budget. The spot is visited using different methods such as short and guided tour etc., and the method needs some quantity of stamina and time and gives a certain level of satisfaction to the tourist. Yin et al. [151], have used Spatial-Temporal LDA (ST-LDA) to come up with an efficient *attribute pruning algorithm* which is fast to recommend the attraction for the tourist.

Kotiloglu et al. [64], have proposed an algorithm that recommend a tourist based on data from the internet sources like social media and various other websites. The authors have used the technique of Collaborative filtering in order to determine few of the attraction that will provide maximum satisfaction to a particular tourist. Chen et al. [22], have used the Chebyshev polynomial approximation method to derive the tourist preferences based on the location and details present for that particular location.

Ying et al. [153] suggested a time-aware metric algorithm that takes into account the user's visiting time at an attraction. In addition, the effect was viewed over a span of time, making this strategy complex. In order to determine the relationship between the user annotations tag and an attraction resemblance, Aliannejadi et al. [2] established a probable model. The results were measured from various LBSN resources and combined with learning to rank strategies. Taking into account the significant time, Zhao et al. [164] suggested a spatial-temporal latent model for suggesting best possible successive attractions. Li et al. [74] suggested to use the 4-dimensional static tensor to store all the data from time and space inspection for the individual visitor, using the Time-aware Factorizing Personalized Markov Chain (TA-FPMC). By measuring the difference between two consecutive tests, the authors took into account the time-decay element. The model was extremely dynamic and all estimates were based on all check-in knowledge. In that work, authors often treated the consumer as a needless extra layer. Therefore, the efficiency of a potential consumer would become diminished if it is used in the recommendation Framework.

The classification of itineraries plays an essential role in understanding the nature of destination. Dietz et al. [30] have proposed a crowd sourcing based *wondary framework*

which is described to schedule and communicate the trips globally. The authors represented an innovative framework such as a content oriented itinerary recommendation to bypass the item detection and drive off problem of the trip model.

Ayala et al. [6] proposed three new approaches such as PHILS, TRILS and SILS to solve the problem of transportation from one POI to other POIs which is not considered by the existing models. The mentioned techniques develop the visit schedules and fine tune it based on the available data. The author did an experiment on Izmir city transport details and its POI data set and found that their recommendation techniques provide dynamic visit schedule with real time data and outperforms baseline algorithms. The average travel duration helps for effective processing and the solution was adjusting the visit schedule appropriately [69]. Laß et al. [69] used different measurement approaches for different schedules. To provide a real value based optimal visit solution along with travel time was the main objective of this work. Only few Internet oriented mobile applications are available for tour trip recommendation of POIs. Already seen POIs, moment of visit, open and closing times of POIs and climate were the main focus of this work.

Lee et al. [70] discussed how the online intelligent tourism data can be helpful for the tourists as well as South Korea's tourism business. Based on big network data analytics, this study concluded that the offline aspirant community had more tourism information rather than online community.

Khatibi et al. [62] studied that the proper forecast on the upcoming incidents is very much needed for the tourism business to attract the tourists and be ready for satisfying them. Many business profits are mainly dependent on the tourism industry capability and its proper predictions of future requirement in every field of the country. Also, this study discussed that, how the social media, tourism website and natural data can help to predict to attract the local and global tourists. Li et al. [72] discussed that due to the Internet, the 'Top-K' best places to visit (BP2V) suggestions are the primary yield factor for online travel groups. However, they need to give more attention to what factors are mainly contributing to finding the 'Top-K' (BP2V). Also had discussed the other related factors like purchasing, foodstuff and enjoyment kinds of characteristics of the POIs.

7.3 Summery and insight

This work also surveys a few works that are specially intended for the TRS domain. Some researchers are concentrating on creating distinct methods that are related to the conduct of travellers. The geo-tagged data is used to extract the tourist information needed. Geo-tagged data enables to acquire the necessary tourist information. Therefore, numerous works have been found, based on geo-tagged data. Few works have been used with a 4-dimensional static tensor to store both time and space check-in information for a specific tourist. However, the efficiency will be degraded if a new user is included in the RS. Therefore, a quick low-level dynamic tensor is also used by the researchers. Itinerary classification plays a key role in knowing the destination's nature. There are few works based on the planning of the itinerary. The researchers are also using AI methods given in Table 12 to solve the tourism-related problem.

7.4 TRS developed based on the AI techniques

There are few techniques available for solving the issues of TRS based on the AI techniques (see Tables 7, 8, 9, 10 and 11). The techniques are given below (Table 12).

Table 7 Summary of Multi-agent systems

Multi-agent systems	Features	System requirements
MARec approach [38] SAMAP [20]	– Recommendations Through Expert and Trusted Agents – Integrates the module which can dynamically capture the models of user – ability to generate attractive tour plans based on the past experiences of the user – Provide services ubiquitously to the citizens and tourist	Random and WTrust FF, IPSS, SIADEX
PaTac [20]	– A personalized travel route is recommended by the agents	RFID, content-based filtering method
TCT ontological recommendation [18]	– Electronic Institutions for regulating the environment also to provide a framework which has been used by Software	ACO, Fuzzy Inference Engine, CDA, TRRA, LTM
Itchy Feet [127]	– 3D virtual world generates a lively community Agent – Used genetic algorithm – Used the problem of traveling salesman	GUI, 3-D graphics, On-line Communities
e-Tourism MAS Architecture [125]	– Agents acts as an expert	Tourism ontology
PersonalTour [85]	– The knowledge about the travel services and stored into its knowledge base and helps to produce an attractive recommendation	Collaboration of multiple agents
Multiagent TRS [19] MARST [9]	– Travel Assistant Agent (T-Agent) developed by graded BDI agent – Reputation Based Collaborative Filtering (RbCF) approach	g-BDI agent JSP, JADE, MySQL 5.0.24

Table 8 Summary of optimization techniques

Optimization techniques	Features	System requirements
TCT ontology recommendation [18]	– Route generation – Route customization	Ant Colony Optimization
PET [40]	– Routing system for the generation of tourist routes that are personalized	MySQL, Apache Tomcat, Google Web Toolkit (GWT)
The City Trip Planner [141]	– Routes for five cities in Belgium – Greedy Randomized Adaptive Search Procedure (GRASP)	Apache web server, JAVA Script, PHP
Context aware tourist app [95]	– Taking a decision about users attraction based on the relevant contextual information	Fuzzy logic, principal component analysis, and Artificial neural networks, C++, GNU GCC, Linux
PSOiA [26]	– Particles are located randomly	
TTDP model [31]	– Particle local and global best position identified	Visual Studio 2015
Managing travel demand [71]	– Helps to plan the route in a new TRS System – Individual decision for traveling – By using the machine learning technique infer unobserved preferences from mobile records	Neural Network approach

Table 9 Summary of automatic clustering

Automatic clustering	Features	System requirements
SAMAP [20]	- Integrates the module which can dynamically capture the models of user - Past experiences of the same users and ability to generates attractive plans	FF, IPSS, SIADEX
Ontology-based feature extraction method [142]	- k-means algorithm for identify the tourist segments set, to identifying the users which have similar demographic characteristics - Identifying the users who have provided their individual ratings	SWOOGLE search engine, Tourism ontology
OWA [148]	Aggregating criteria guided by quantifiers	fuzzy logic
Hybrid context aware system [36]	recommendation system uses the fuzzy c- means as well as k-means uncertain version	Fuzzy data analysis
A hybrid recommendation approach [89]	each cluster and for each object, the membership lies between the value of 0 and 1. For each group the summation of membership degrees equals 1	Fuzzy c-means
Tourist@ [8]	- Classification based on association is applied - Clustering has been applied all the time that 10 users can join newly	Content-based and Collaborative recommendation strategies
ClusDM [139]	periodically it is require to recompute the classes	Lisp and Java
SPETA [43]	- Solves multi-criteria decision problems - Uses the knowledge of the user's present location, preferences and the user's past locations. - suggested a TRS by using the Support Vector Machines (SVMs) where the different user preferences are stored into the vector.	GIS modul
W2Go [39]	- Travel based on the landmark ranking - Developed using the statistics of photo from Flickr and also the knowledge of the user from the Yahoo travel guide	Normalized Cuts algorithm, Flickr API
Community-based Social Tag Clustering [108]	- Automatic clustering of Social Tags (ST) with the help of community detection. - Identified that the ST network has scale in free properties	Seekda search engine, Web Services Description Language (WSDL)

Table 10 Summary of Management of Uncertainty

Management of Uncertainty	Features	System requirements
ITAS [53]	– Possible number of attributes influence directly on the probability where the users have an certain tourist point of interest – Integration of EBM model, Bayesian network	Windows 2008 Server IIS7.0, Visual Studio 2008, Netica 4.09, Google Maps API
Online personalized RS which is enabled semantically for planning the tour [53]	– linkage about the users when users linking a certain tourist destination – Heterogeneous integration, personalized recommendation, adaptive capability, spatial functions capability	Java/Servlet, JSP, ArcWeb service, Netica-J 2.2.1
Context aware tourist app [96]	– Taking a decision about users attraction based on the relevant contextual information	Fuzzy logic, principal component analysis, and Artificial neural networks
Visitor Information System [67]	– rule-based approach wherein a system has certain rules which generate recommended events based on the user preferences.	Mobile services, Contextual Computing

Table 11 Summary of Knowledge representation

Knowledge Representation	Features	System requirements
3D-GIS Hybrid RS [124] SMARTMUSEUM [118]	– An interface based on 3D map and real-time location-aware recommendation – Ubiquitous RS for the Web of Data – Bridges the semantic gap that exists in the sensor inputs, heterogeneous content descriptions and the user profiles. Ontologies are used in this case. – User preferences are stored into each node of the ontology – minimizes the cost for constructing the ontology – Specific types of services are accessed – Semantically enabled e-Tourism Portal – collects data on the basis of semantic ontologies	GPRS and UMTS, C++, GNU/Linux Kappa statistics
SigTur/E-Destination [13] Ontology Population [117] Ontology-based Framework [27] OASIS [45]		Collaborative filtering Named entity recognition, Ontology Population phase Web Server 9.0, JSP, Enterprise Java Beans (EJB) DataLift, CSV, JSON

Table 12 Summary of recommendation methods using RS, TRS and AI Techniques

RS	TRS	AI Techniques
CBRS: (L et al., 2009)-(Huang & Bian, 2009b) CFRS: (Savir et al., 2013)-(Yang & Hwang, 2013) KBRS: (Ruotsalo et al., 2013), (Borrás et al., 2012), (Sebastia et al., 2009b), (Ruiz-Martínez et al., 2011), (Daramola et al., 2009), (Gasmí et al., 2016) HRS: (Shambour & Lu, 2011), (Umanets et al., 2014), (Burke, 2002), (Al-Shamri & Baradwaj, 2008), (Lucas et al., 2013), (Fenza et al., 2011), (Sebastia et al., 2009b), (Garcia et al., 2011) CIRS: (Yu et al., 2017), (Banerjee et al., 2010) SNRS: (Jannach et al., 2010), (Ricci et al., 2010), (Umanets et al., 2014), (Seidel et al., 2010), (Garcia-Crespo et al., 2010), (Pearl, 1988), (Jamali & Ester, 2009), (Yang et al., 2012), (Umanets et al., 2014), (Yu et al., 2017) CARS: (Hsu et al., 2007), (Dey et al., 2001), (Zheng & Xie, 2011), (Lucas et al., 2013), (Lamsfus et al., 2009), (Lamsfus et al., 2011), (Wang et al., 2011), (Gavalas & Kenteris, 2011) GRS: (Zheng & Xie, 2011), (Garcia & Sebastia, 2014), (Garcia et al., 2011)	(MinFeng & Athanasopoulos, 2011), (Lim et al., 2015), (Yin et al., 2011), (Ennis et al., 2013), (Liu et al., 2014), (Ickjai et al., 2013), (T-h & S-B, 2016), (Yuan et al., 2016), (Majid et al., 2015), (De C et al., 2014), (Cheng et al., 2011), (Clements et al., 2011), (Brilhante et al., 2013), (Brilhante et al., 2015), (Lu et al., 2011), (Souffriau et al., 2008), (E-y et al., 2006b), (Lim, 2016a), (Kurashima et al., 2013), (Lim et al., 2016), (Maruyama et al., 2004), (Yu et al., 2015), (Zheng & Xie, 2011), (Yuan et al., 2012), (Uddin & Ravishankar, 2011), (Yoon et al., 2010), (Lu et al., 2010), (De C et al., 2014), (Yu et al., 2016), (Kotiloglu et al., 2017), (Altanajedj & Crestan, 2018), (Li et al., 2017), (Liao et al., 2018), (Dietz & Weimert, 2018), (Herzog et al., 2018), (Brudy et al., 2014), (Mottini et al., 2018), (Mottini et al., 2018), (Kapak et al., 2018), (Li et al., 2019)	MAS: (Gao et al., 2018), (Castillo et al., 2008), (C-s et al., 2009), (Seidel et al., 2010), (Sebastia et al., 2010a), (Lorenzi et al., 2011b), (Casali et al., 2008), (Bedi et al., 2014) Optimization techniques (C-s et al., 2009), (Garcia & Arbelaitz, 2010), (Vansteenvogen & V, 2011), (Meehan et al., 2013a), (Dang et al., 2013), (Ding et al., 2016), (Leng et al., 2016) Automatic Clustering: (Castillo et al., 2008), (Vicent et al., 2013), (Yager, 1988), (Fenza et al., 2011), (Lucas et al., 2013), (Bate et al., 2012), (Valls, 2003), (Garcia-Crespo et al., 2009), (Gao et al., 2010), (Pan et al., 2013) Management of Uncertainty: (Huang & Bian, 2009a), (Meehan et al., 2013b), (Lamsfus et al., 2009) KR: (Sebastia et al., 2009b), (Ruotsalo et al., 2013), (Borrás et al., 2012), (Ruiz-Martínez et al., 2011), (Daramola et al., 2009), (Gasmí et al., 2016)

7.4.1 Multi-agent systems (MAS)

An agent is basically a computer system which has the ability to take an independent actions on behalf of its particular owner or user [18]. On the other hand in case of MAS, multiple agents work together for sharing their information, their activities, resources etc. and producing a joint efficient outcome of a distributed problem (see Fig. 5).

Casali et al. [19] proposed a TRS which consists of multi-agent architecture where the Travel Assistant Agent (T-Agent) is developed by graded BDI agent (see Fig. 9). Castillo et al. [20] proposed a software tool named SAMAP which basically integrates the module which can dynamically capture the models of the user, list of activities and subsequently

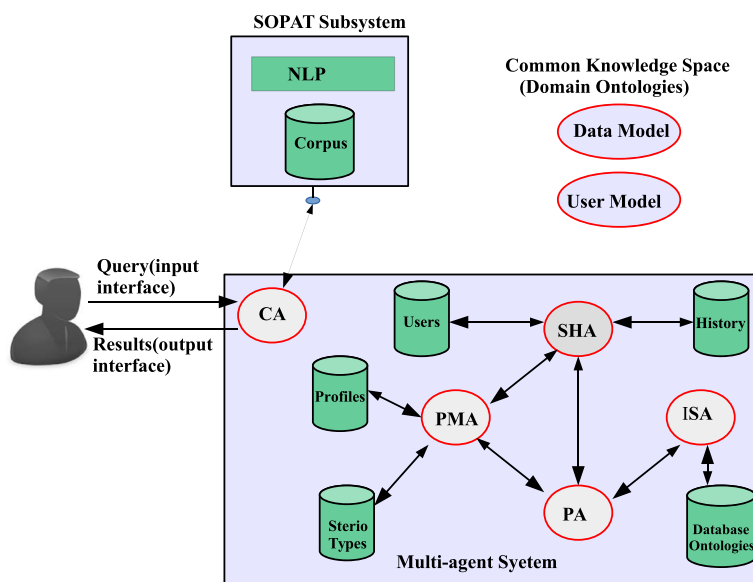


Fig. 9 Functional, MAS architecture [21]

generate attractive plans for the user. SAMAP is basically based on MAS and consisting of user modeling, Case-Based Reasoning (CBR), interface agent and planning agent. Ceccaroni et al. [21] proposed a TRS named as PaTac is basically consists of three main component, namely the Content Collector (CC), agents, and Content Deliverer (CD). The agents are responsible for storing the necessary information in ontologies and the CD provide the services to the other component. The PaTac technique consists of different components like Communication Agent (CA), Session Handling Agent (SHA), Profile Managing Agent (PMA), Personalization Agent (PA), and Information Service Agent (ISA). Initially, CA receives the query, encloses the Agent Communication Language (ACL) message and subsequently submits to the SHA. The SHA is responsible for handling the user session and through the PMA it initializes the U_p and finally submits the PA user request. PA is responsible to gather the preferences of the users via PMA and it also gathers the content from the database and ontologies via ISA. Ultimately, by using the personalization algorithm PA generates the result for the particular user. C-s et al. [18] proposed a technique for Tainan City Travel (TCT) (see Fig. 6). The authors have used the concept of Travel Route Recommendation Agent (TRRA) and Context Decision Agent (CDA). Initially, a domain expert produces the ontology of Tainan City travel. A Location Transfer Mechanism (LTM) has been used, which depends on the information received from TCT and is used to generate the location value, which is subsequently saved into the ontology. TCT ontology is retrieved by CDA in order to determine the *tourist matched concept* and then computes the semantic as well as the context relation for each instance. On the basis of the output of CDA, a fuzzy inference engine is constructed by TRRA to infer the context information. Sebastia et al. [125] proposed a multi-agent architecture for TRS. The authors have proposed a system which integrates various agents, helps to dynamically collects the U_p and also to analyze the user activities with the help of experience from the user. Lorenzi et al. [85] came up with a recommendation technique on the basis of multi-agent platform named as “PersonalTour”. When a tourist request for recommendation, the agents acts as an expert and provides an attractive recommendation. Lorenzi et al. [84] proposed a RS based on the trust and the expert agent. To bring down the communication load and to increase the quality of information exchanged between the agents, authors have considered the communication from the trusted sources. Seidel et al. [127] proposed a technique for TRS which is built on the 3D e-tourism recommendation technique named as Itchy Feet. The authors have considered the Electronic Institutions for regulating the environment and have also provided a framework which has been used by the Software Agent (SA). The SA provides information from the environment where social interactions are implicitly addressed and the 3D virtual world generates a lively community. Bedi et al. [9] suggested a TRS technique based on the Reputation based Collaborative Filtering (RbCF) approach named as MARST. MARST consist of various agents and has been described as follows:

- Specialized agents (Data Repository Agent (DRA)): the necessary information are extracted or aggregated from the web services.
- Items Reputation Agent (IRA): calculates the reputation of all the items.
- Similarity Agent (SA): calculates the similarity between the users.
- System Agents (User Agent (UA)): helps to interact with other agents. UA receives the user query request and transfers it to either the HRA, RRA or PRA.
- Hotel Recommendation Agent (HRA): it takes the query from the UA
- Place Recommendation Agent (PRA) and Restaurant Recommendation Agent (RRA): Service recommendation is computed for both the cases.

7.4.2 Optimization techniques

There are several optimization techniques that are used for optimizing TRS. C-s et al. [18] proposed *Ant Colony Optimization (ACO)* technique for finding the personalized travel route (see Fig. 10). Garcia et al. [40] proposed a routing system for the generation of tourist routes that are personalized. Vansteenwegen et al. [141] proposed an Expert System (ES) based on the Greedy Randomized Adaptive Search Procedure (GRASP) algorithm. Meehan et al. [95] developed an intelligent RS by accessing the user's context using Artificial Neural Network (ANN). D-c et al. [25] used PSO-inspired Algorithm(PSOiA) for solving the TOP problem. In PSOiA, at the very beginning, the particles are located randomly and then iteratively location of each particle is replaced by new one using Randomized Heuristic Algorithm (RHA). The *local search algorithm* is used to determine the local and global best position for every single particle [26]. Ding et al. [31] proposed a TRS using route planning based on Fireworks Algorithm [136]. The authors have considered the Tourist Trip Design Problem (TTDP), which is basically a discrete optimization technique. Leng et al. [71] came up with a technique which does location based recommendation depending on the Geo location data captured by the mobile phone. The authors have used optimization technique for improving the efficiency of a system.

7.4.3 Automatic clustering

Clustering is defined by the act of partitioning the dataset which is unlabeled into groups of objects which are similar [133]. The objects of the same group are similar but dissimilar with other group's object. There are many recommendation systems which has been developed using collaborative filtering, where the users are partitioned into groups and user of each group share similar characteristics. On the basis of demographic information, user individual ratings of each item and user preferences, the similarity of group users has been identified. Castillo et al. [20] identified the similarity of users on the basis of a domain ontology. Here the system concludes that, a user preferring Cinema is more similar to users who

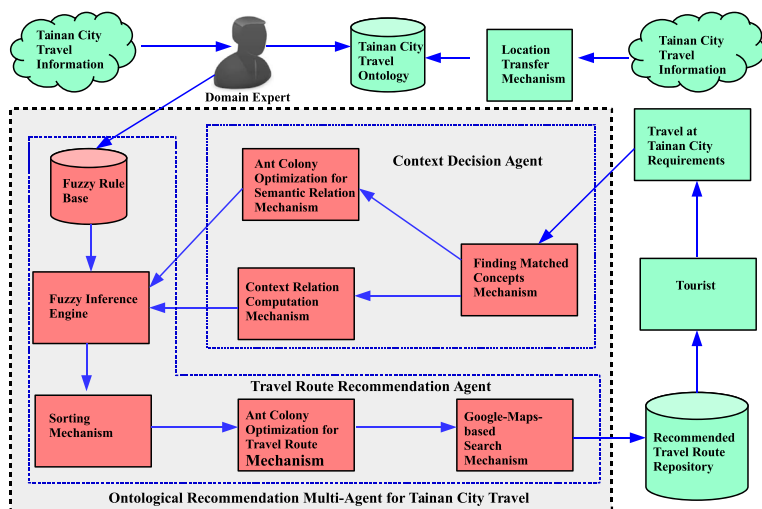


Fig. 10 Ontological recommendation multi-agent Structure for Tainan City travel [18]

enjoy the Theatre than any Spots activities. Vicient et al. [142], used the k-means clustering algorithm to identify different tourist segments having similar demographic characteristics and individual ratings. Yager et al. [148], and Dujmovic et al. [32] used OWA and LSP to add similarities based on users' ratings using the well-known Pearson correlation as a similarity measure.

Fenza et al. [36], proposed a RS using the fuzzy c-means as well as k-means uncertain version. The authors have defined that for each cluster and for each object, the membership lies between the values 0 and 1 and for each group the summation of the membership is equals 1. Lucas et al. [89] proposed Personalized Sightseeing Information System (PSIS) using same technique [36]. Batet et al. [8], proposed a system wherein the clusters are formed every time when a group of ten new users join the system. The underlying principal states that the addition of users requires the classes to be recomputed periodically. Valls et al. [139] proposed a clustering system in which a hierarchy of different classes has been created on the basis of the user's interest.

Gao et al. [39] developed a system for the proper guidance of travel based on the landmark ranking. The system has been developed using the statistics of photo from Flickr and also the knowledge of the user from the guide present at Yahoo! Travel. Pan et al. [108] developed a system on automatic clustering of Social Tags (ST) with the help of community detection. Authors have identified that the ST network has scale in free properties. Garcia-Crespo et al. [43] suggested a TRS by using Support Vector Machines (SVMs) where different user preferences are stored into the vector. To calculate the gap between the preferences of the user and the recommendable items, authors have considered SVMs for identifying the most appropriate ones.

7.4.4 Management of uncertainty

Developing an intelligent RS by fully understanding the requirements of the tourist properly is a challenging task. The demand of each tourist may be different and sometimes almost same. A lot of works has been done towards developing an understanding the tourist activities based on AI techniques. Pearl et al. [111] used Bayesian network which is based on acyclic graphs, where the edges of the graph represents relationships of causality between available nodes. Later, Hsu et al. [50] used Bayesian network where the possible number of attributes influence directly on the probability where the users have an certain tourist point of interest. Huang et al. [53] proposed a network, wherein the attributes are influenced by the type of the user and also there exists linkage between the users and the tourist destination. To handle the uncertainty of tourist there are some works which have been developed based on the Fuzzy logic. A series of linguistic labels have been taken by Fuzzy variable and also each label is considered a Fuzzy set where the membership function lies between 0 and 1. There are many users whose preferences can be represented using the Fuzzy sets and fuzzy reasoning to calculates the user's preferences, [18, 43]. A Restaurant can be stated as either good or bad and is represented by the value between 0 and 1 but it is also possible that for further analysis good or bad comments may instead be given, 2013 [96]. Lamsfus et al. [67, 68] developed a RS based on the rule-based approach where in a system there are certain rules which generates recommended events based on user preferences.

7.4.5 Knowledge representation (KR)

The clear information about the domain is called as the domain knowledge. The main issue is to gather full knowledge from the domain, because it comes from different sources and

also should be integrated. For designing a knowledge base, individual and the relationship should be clearly represented. Knowledge is required to solve any problem as well as to design the program. AI can be used in a satisfactory manner for this purpose. Presently, the most common method for representation of the domain knowledge is through ontologies, which helps to describe clearly and the ideas about the domain. The ontologies store necessary information about different features which are used for generating an intelligent RS. Wang et al. [143] considers a user and travel ontology where tourists' demographic characteristics as well as their preferences are modeled. [20, 91], and Ruotsalo et al. [118], Alonso et al. [3] have also used ontologies for designing the activities of different tourists. The authors used ontology-based similarity for calculating similar activities of different tourists and providing recommendation to the tourist based on the technique of collaborative filtering.

Borràs et al. [13], Moreno et al. [103], Sebastia et al. [124]; [126]; Garcia et al. [42], have also used ontologies in their system. The user preferences are stored in each node of the ontology [13, 103] and the values are generated by the users who have filled the questionnaire about their experiences and preferences. Borràs et al. [13] proposed ontology-based system to dynamically manage the user preferences.

Castillo et al. [20] proposed PaTac which uses different ontologies of restaurants, entertainment, sports etc., and to integrate them authors have used standard temporal and geo-location ontologies given by the World Wide Web Consortium (W3C). Lamsfus et al. [67] proposed a system called CONCERT, wherein all the travel context is stored in an ontology network called Contology. Ruíz-Martínez et al. [117], and Vicient et al. [142] came up with a technique in order to minimize the cost for constructing the ontology. Xu-yin et al. [147] suggested a new approach on intelligent RS based on the WebGIS by applying the knowledge representation theories of AI. Daramola et al. [27] suggested a TRS framework which is based on ontology. The framework is divided into three layers, client-tier where specific types of services are accessed, middle layer is called Semantically enabled e-Tourism Portal and third layer is the layer of ontology such as destination, travel, accommodation, Restaurants and event ontology respectively. The architecture of this framework has been shown in Fig. 11. Umanets et al. [138] proposed a RS based on the user current location named as SPETA. Gasmi et al. [45] proposed an approach for TRS named as OAI-SIS. The authors have provide the solution of data management based on the CrowdSensing technique.

7.5 Summery and insight

Most of the techniques designed for TRS are either a client-server or a stand-alone system where there is no opportunity to exchange information among agents. Many works have been developed on the basis of MAS to solve this problem. On the contrary, there are several optimization techniques that are used for optimizing TRS. The authors have used optimization technique for improving the efficiency of a system. There are many recommendation systems that have been created using clustering techniques, where tourists are divided into groups and tourists of each group share comparable features.

Figure 12 shows that most of the existing works are using RS methods to solve the TRS problem followed by the techniques specially designed for TRS and AI methods. As discussed in Section 3.1, there are various factors which are considered for recommendation in TRS. In the literature authors have used these factors either singly or jointly. Figure 13 shows the distribution of existing works with respect to different factors. As discussed in

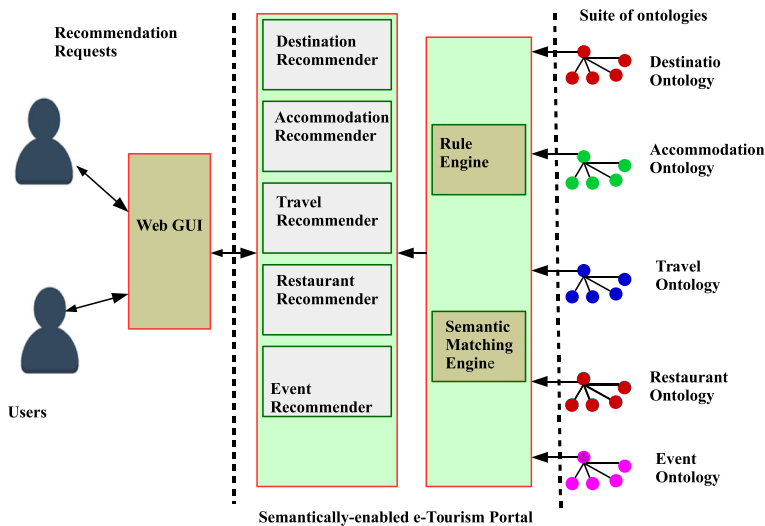


Fig. 11 Ontology-based Architecture Framework

Section 3.1 TRS suggests different features to the tourist. Figure 14 shows the distribution of existing works based on the features they suggest.

8 Future Research directions

For developing an intelligent RS, there exists few research challenges which have been described as follows.

- Currently most of the RS are focusing on the representational view of the context. However, it is very important to know the various context knowledge such as partial and static, partial and dynamic, latent knowledge of context and also unknown context. In such cases an intelligent system is required which can work with all these factors and provide the best RS to the tourist.

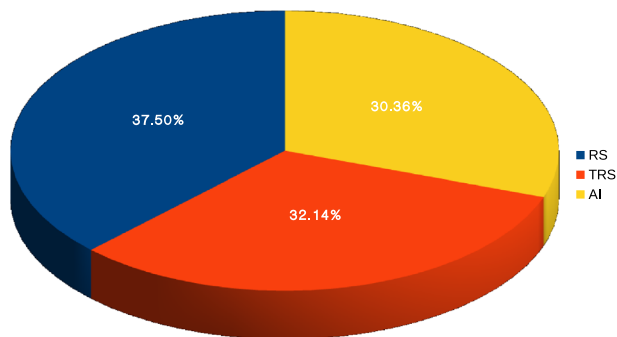


Fig. 12 Tourism Recommendation system: Based on RS, TRS and artificial intelligence

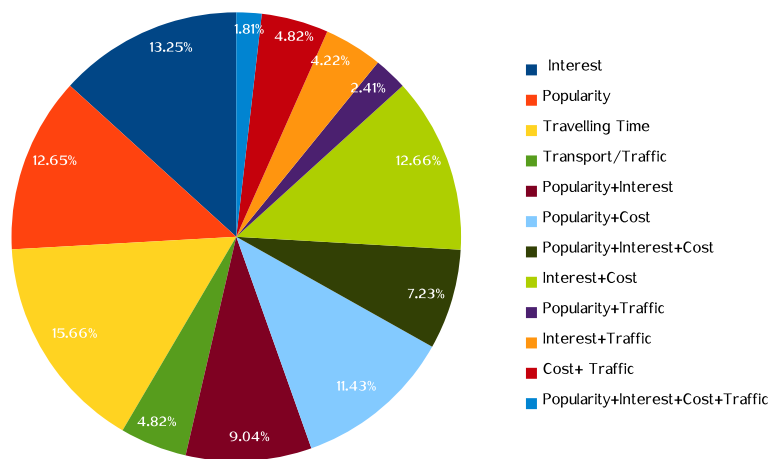


Fig. 13 Tourism Recommendation using tour interest, popularity, travelling time and traffic condition

- Most of the existing works mentioned by previous researchers are based on prefiltering method , postfiltering method as well as contextual modeling method. However there is a need to explore alternative methods in a comprehensive manner and also to identifying the limitations of these methods.
- During the clustering process for a large scale implementation it may be difficult to derive the satisfaction functions.
- Sometimes when an item is shown to the user, instead the user follows the other item which clearly shows that the user has more interest in the other item than the current item at hand. So by designing a proper user interface (UI) an individual satisfaction can be increased.
- Most of the work considers the accurate profile of user’s individual preferences but in real scenario the data are probabilistic and also most of the work tries to solve this

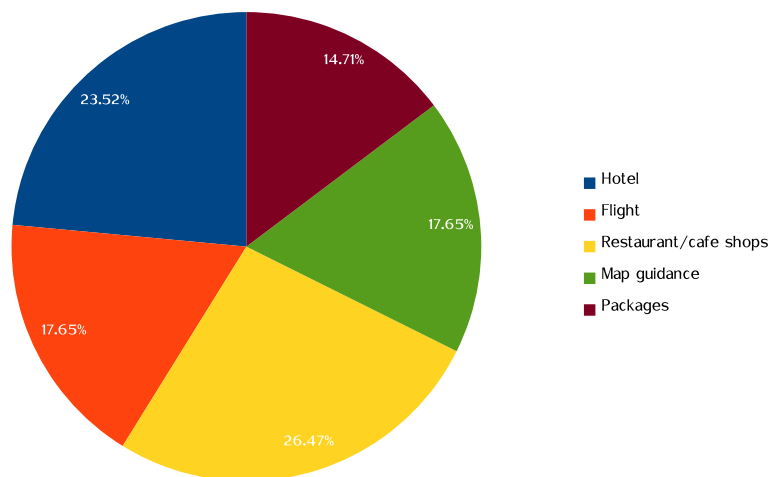


Fig. 14 Tourism Recommendation using different features

uncertainty based on the Bayesian networks but it is required to focus more on the average and plurality voting strategies.

- Privacy preserving is a great challenge to designing a RS system as it requires user data to make individual recommendations.
- The POI sequence recommender is based on some criteria like time, individual priority and budget and its score. The score is calculated based on individual priority, trip duration between POIs and POI popularity. Hence, there is a need for an efficient optimizing approach which increases the score of the related criteria.
- The tourist preference may overlap with time. So an itinerary gets the most importance rather tourist's interest preferences.
- The future work has opportunities like to consider the date, time and climate of the location for the best recommendation to satisfy the user.
- Also, the future work has the scope like merging of POI popularity and tourist interest with mobile application and provide timely POIs recommendation.
- If a user does not have any idea about the target place, in such cases there should be a certain degree of diversity amongst all the items which are there in the recommendation list.
- Sometimes there are large number of items available for the users and in such cases a user may give ratings to few items only.
- The mobility of users poses great challenge and the user may need recommendation at the new locations. In such cases it is very important to know the places that the user has already visited along with his current location.
- To avoid wasting and waiting time of single transport mode during shifting process, the future research direction has a scope of various form of transport suggestion like flight, taxi, bus, etc. in itinerary recommendation along with merging of diverse groups like family, youngsters, seniors, etc.
- To design a dynamic itinerary recommendation approach is a great challenge which can overcome the loss of POI coverage due to environment factors like rain, flood, hot climate, etc. The algorithm may be designed to suggest alternative safe POIs to visit in such kind of environment.
- Same type of political, tradition, language itinerary suggestion and low cost as well as visa-free countries, itinerary recommendation based on tourists interest is also a good choice of future research direction.
- To recommend new user itinerary based on the old user feedback such as their fulfillment/disappointment ratings, time spent, interest, etc. of POIs, will enhance the itinerary recommendation process.

9 Summary and conclusions

This work discusses the salient points about TRS and its importance to the tourists to plan their trip in the best possible manner. Tourist data can be extracted from various social media platforms and can be used further in TRS. Specific techniques which are targeted towards TRS have been discussed in detail which clearly show us that the tourism domain is unique in nature and TRS needs to handle the various aspect like POI, Itinerary, etc., in a special manner. Several examples of TRS have been highlighted along with their salient features. Also, future research direction has been suggested to motivate the young researchers.

Declarations

Conflict of Interests The authors have declared that there is no conflict of interests exist.

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